ENERGY SECURITY
Overcoming the energy crisis in the country

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Energy Security clearly has to be South Africa's priority, given the enormous cost of loadshedding to the economy and to welfare.

That doesn't mean that the other two dimension of the energy trilemma – namely affordable energy and environmental sustainability – are irrelevant. As shall become clear, our future energy path is one where all three policy objectives – clean energy, competitive prices and security can be achieved because the least-cost energy path is also one which involves climate-friendly technologies and where an adequate and reliable supply can be achieved.

The contrast of our current situation to these laudable goals and outcomes could not be starker. We have zero energy security with almost daily power cuts. Electricity prices have increased nearly 6oo% in nominal terms over the past 15 years. And we have the most carbon-intensive power sector in the word (expressed as carbon dioxide emitted per kWh produced), which will likely begin to threaten the competitiveness of our exports through the imposition of the carbon border adjustment and other mechanisms.

But back to electricity security. What are the prospects of load-shedding or power cuts ending soon? In answering this question, I first want to deal with a number of myths or false narratives; second, I'll assess current initiatives; and then third, I'll refer to the energy transition and how it can help us restore electricity security.

The first myth or false hope is that Eskom sits with 15,000 MW of idle generation capacity that is not being used and that all that is needed is to bring this back online – as energy Minister Mantashe has said on a number of occasions.

Of course, that statement ignores that that this kit is broken, and continues to break, and that for more than 12 years now, successive Eskom management teams, supported by their technical staff, have tried their best to reduce these unplanned capacity loss factors, but the energy availability factor (EAF) of Eskom power stations has continued to decline, from above 90% in the early 2000s to below 60% today, and at times even below 50%.

Comments on idle capacity, and even the perennial expressions of hope that Eskom coal power station performance can or will soon improve significantly, are irresponsible, fact free, and fly in the face of actual performance data over time, and mostly do not engage in any useful detail on what Eskom should be doing, that has not already been tried, to reverse the situation.

A fundamental step in solving South Africa’s loadshedding crisis is a recognition and acceptance that it is very difficult, and very expensive, to reverse Eskom’s long-term decline in the electricity available from its coal fleet. Eskom does not have the reserve margin or space to take sufficient plant offline to fully catch up on maintenance, especially for lengthy refurbishments. It does not have sufficient free cash flows to fully fund the required maintenance. And it has lost far too much of the experience and skills to undertake effective, comprehensive maintenance of the scale and quality required. Most commentators on Eskom completely underestimate the complexity and gravity of these challenges.

Yes – there have been some recent gains. There is a greater openness to accepting private sector advice. Some experienced grey beards have been brought back, although not nearly enough. National Treasury’s debt relief package has meant that Eskom now spends less repaying principal and interest on its loans and more cash is potentially available for maintenance. Procurement processes for parts and engaging Original Equipment Manufacturers have been streamlined. Incidents of sabotage have reduced.

Six power stations are being focused on and have been targeted for major improvements. But weren’t they also in the past? Much is being made of the fact that newly appointed generation and power station managers are making a difference. But overall, Eskom governance is in disarray with the recent resignations of the Chair of the Board and no permanent Chief Executive. Indeed, Eskom is being run by its CFO, who has indicated that he wants to leave.

Electricity Minister Ramokgoba has made a real contribution to increasing the transparency and regularity of information flows on what is being done to reduce Eskom loadshedding. Politically, he is under immense pressure to show results and often that unfortunately leads to optimistic interpretations of the data. Yes – there will be times when things improve but data on the long-term deterioration in Eskom performance is clear. I can guarantee you that there will be more plant breakages and that there will once again be periods of severe loadshedding in the short to medium term.

The second myth is that renewable energy cannot make a difference to load-shedding. The sun doesn't always shine, or the wind doesn't always blow, so how can they make a difference? Again, this is a false narrative that is fact free. Eskom's own daily data releases show that wind energy alone frequently contributes more than 2000MW, importantly also helping to meet peak demand. In other words, if those wind farms were not there, we would be having up to two stages of worse loadshedding at times. And similarly with solar.

But here is some solid science that shows the potential impact of solar and wind energy in reducing loadshedding. As an exercise, hourly Eskom data over a year was taken for electricity demand and supply and loadshedding, and the impact was modeled of adding 5000MW extra solar and wind energy, producing in exactly the same profile as existing solar and wind plant. So, these are not theoretical, ivory tower academic assumptions. The analysis is based on actual Eskom data. The results are truly extraordinary. If we had done this - i.e. added an extra 5000MW - in 2021 there would have been 96% less load shedding, and last year 70% less.

How is this possible, you might say. Solar and wind are intermittent, variable. Well, the answer is twofold. First, there is the direct effect: in the context of ongoing loadshedding, any new electrons on the system make a difference and reduce loadshedding. But, second, there is an important indirect effect. Extra energy on the system creates the space for replenishing diesel and pumped storage. Much of our current loadshedding is when these resources have been overused and when Eskom does not have enough energy or space to replenish storage.

I'm using the examples of solar and wind energy because it has now been proved, in the auctions run by the IPP Office, that they are the cheapest new sources of energy - around 50c/kWh compared to Eskom's average cost of supply which now exceeds R1.40 c/kWh.

And this highlights the third false narrative, namely that renewables are expensive. Yes – they were, a decade ago, but no longer. The costs of solar PV fell by 90% and for wind energy 70%. We're interested in the future, i.e. the cost of the next kWh to be added to the grid.

Minister Mantashe has struggled with all of these myths and continues to argue that Eskom has idle capacity, solar and wind cannot make a difference and are expensive. The consequence of these false beliefs has meant that not a single kWh of electricity has yet been added to the grid commercially from any project that he has procured in the more than four years he has been energy minister, despite him having extraordinary powers under the Electricity Regulation Act to determine how much electricity, from which sources, should be procured by whom and by when. This is a devastating indictment, as the economy reels from loadshedding.

And so, we cannot restore energy or electricity security purely by thinking we can fix Eskom. We need also to add new power generation.

There is a fundamental conceptual issue here which has huge strategic and investment implications. Eskom has been failing at a rate faster than we have added new capacity. That departure point has to be accepted in order to solve load-shedding.

We might wish it were different. We may wish to hold on to the belief that Eskom can be a key instrument in the developmental state when patently it is not – it's crippling the economy and draining the fiscus. But we have to deal with reality and act on that, including acceptance that more and more Eskom power stations will begin exceeding their design life of 50 years – indeed Eskom envisaged that 11GW would need to be retired by 2030 and a total of 22GW by 2035. That programme may be slowed, but the fact is that South Africa will continue to lose Eskom generation capacity.

And Eskom itself estimates we need 50-60GW of new solar and wind energy by 2030, complemented by storage, gas and new grids, amounting to investments totaling R1.2 trillion

Electricity Minister Ramokgoba has bravely acknowledged these facts and was reported this week as saying: “ An overreliance on these [Eskom coal] units I think is particularly unhelpful; it’s the height of folly to suggest that, going into the future, this is going to resolve our problems . . . and that’s why we must accelerate the onboarding of new generation”.

So what can we say around government’s response to our power crisis. Clearly, it has not done enough. We had loadshedding in 2007 & 8, 2014 & 15 and every year since 2018, and every year it has got worse.

Nevertheless, the President’s Energy Action Plan, announced in July last year, is broadly well conceived and framed. It comprises three broad goals and desired outcomes: first, every effort needs to be made to at least stabilise Eskom (we rely still for 85% of electricity from the utility); second, investment in new private generation needs to be accelerated; and third, the power sector needs to be restructured sustainably.

The National Electricity Crisis Committee, supported by Business, is making steady progress in these three areas. I’m not going to report on the detailed work that is being done but let me make an overall assessment.

While the interventions at Eskom are positive, I think there is still not sufficient appreciation of the complexity of turning Eskom around and rebuilding the knowledge, skills, systems, processes, practices and budgets to make a step change in the performance of old, long-neglected plant.

And we are moving far too slowly on new power generation. Again, I don’t believe there is sufficient appreciation of the scale of what is require: more than one trillion rand, 60GW by 2030. We are way off those targets.

While we are seeing encouraging signs of private investment in generation – for example Eskom estimates close to 5GW private solar generators have recently been added (these are mainly rooftop and non-REIPPP investments) there are still major obstacles in connecting new power, not least transmission constraints.

And, we have to admit now, the IPP Office, under the DMRE, is a failure. A hard thing to say, I know, but let’s look at the facts. Between 2011-14 the IPP Office ran four auctions that delivered 6300MW of solar and wind projects that are now all delivering electrons to the grid. Then Messrs Molefe and Koko, with the support of President Zuma, blocked the programme, the contract of the well regarded and internationally lauded head of the IPP Office was terminated, and frankly we now have a team there that have failed to deliver.

The so-called emergency procurement has so far delivered a measly 150MW (a solar and battery installation that is currently being commissioned). Bid windows 5 and 6 have not yet delivered a single electron. Bid window 7 seems to be disarray. The battery and gas bid windows keep on being delayed.

Within the context of severe loadshedding, and its cost to the economy, this is completely unacceptable. In addition to private sector initiatives, we need to keep these public procurement initiatives going. International experience demonstrates that centrally run auctions deliver big megawatts at competitive prices.

South Africa has the capabilities to do this well. The world applauded our Bid windows 1, 2, 3 and 4. We need to bring those skills and capabilities back into the IPP Office, which needs to be removed from the DMRE and institutionally located in the new Central Purchasing Agency of the Transmission System Operator (TSO).

Which brings me to Eskom restructuring. Again, progress has been far, far too slow. The Energy Policy White Paper of 1998 said we would separate out transmission from Eskom. When I chaired President Ramaphosa's Eskom Sustainability Task Team in late 2018 and early 2019, we resurrected these proposals. The president accepted our recommendations and announced in Parliament that Eskom would be split into three, starting with Transmission. Four years later, Minister Gordhan as not yet fully implemented this decision. Now there is a mad scramble to get the Electricity Regulation Act Amendment Bill through this parliament, which will enable the establishment of the TSO.

These restructuring proposals may seem arcane and not terribly important, but they are fundamental. A Transmission System Operator, separate from Eskom, removes Eskom's conflict of interest as both a generator and also the owner of the monopoly transmission grid. It creates a fair and transparent platform for contracting and trading with multiple generators, thus accelerating private investment and competition.

And, critically, an independent TSO will be insulated form Eskom's financial woes. It will generate predictable, regulated revenues, start migrating back to investment grade, thus accessing competitively priced capital markets, and investing in strengthening and expanding the grid to connect new power generation.

This could have happened already, if the decisions of the President and Cabinet had been implemented. Even the simple step of appointing a board for Eskom's transmission subsidiary has not been done, despite excellent and knowledgeable candidates being identified in the search and interview process. Doing nothing has consequences.

During the parliamentary consultation on the ERA Amendment Bill, concerns have been expressed that the creation of the TSO implies privatisation. We need to explain that categorically it does not. Privatisation means the sale of state assets. The TSO will remain publicly owned and will be a state-owned enterprise.

Let me end with the reminder that we're in the midst of a radical energy transition. I often say to my students that I have seen more change and innovation in the power sector in past five years than in the previous 35 years of my professional career. Disruptive innovation in enabling technologies and business models is having a major impact on power markets, system operation and regulation.

These innovations include the break-through of unsubsidized solar and wind which are now the cheapest sources of new power. As their share on the grid grows, they need to be complemented by flexible resources such as hydro, pumped storage (operated in innovative ways), batteries and also gas.

Balancing and capacity markets will need refining, including the competitive procurement of flexible ancillary services to maintain system voltage, frequency, strength and inertia. The capabilities of system operators, including at the regional level, will need to be enhanced.

Demand-side resources and virtual power stations will be critical parts of the system.

The distributed nature of renewable energy means distributed energy resources will multiply with implications for transmission and distribution planning, business models, and distribution system operation. City utilities will once again become key elements of the power system.

More electricity consumers will become producers as well (i.e. prosumers). As power flows both ways, advanced net metering and payment systems will provide greater spatial and temporal granularity. The internet of things, artificial intelligence, big data, and blockchain, mobile money and pay-as-you-go models will change the way in which electricity is controlled and paid for. Peer to peer trading between prosumers will begin, also across connected mini-grids. There will be more aggregators and traders in the market. Community-ownership models could arise alongside utilities.

All this means that we need informed and far-sighted leadership in the sector, especially in managing the energy transition which unfortunately is increasingly contested. Of course, some constituencies will be adversely impacted by the energy transition and we need to ensure that it is just – that workers and communities in the coal belt are supported through re-skilling and focused investments.

Again, I need to call out responsible Ministers. It is irresponsible to continually questioning the role of clean energy technologies and the just energy transition. And once again we need to challenge false narratives. It is now almost a mantra that Germany tells us to go green but has massively increased it imports of our coal. While it is true that the disruption of Russian gas supplies resulted in Germany temporarily using more coal, the data now clearly shows that European imports of South African coal have declined, as has electricity produced from coal. In the first 9 months of 2023, coal contributed 26% of Germany's electricity, while renewable energy contributed 60%. Facts. Not myths.

Indeed, is amazing that South Africa has placed so many obstacles in liberating the US$ 8.5 billion just energy transition fund promised at COP26. I have spoken to senior people in these development finance institutions who have expressed frustration that despite being willing and ready to disburse these funds they are unable to get any traction or make any progress with the South African government.

We know that a large chunk of this money could be directed to strengthening and expanding our transmission system that would connect new power generation to alleviate load shedding. Yet, individuals choose to criticize these funding opportunities as a new kind of colonialism.

In conclusion, South Africa can end loadshedding and restore electricity supply security but it's going to take time. There may be short periods of respite but loadshedding will return as old Eskom generation plant continues to deteriorate and older plant exceed their design lives.

We need to massively scale up investment in power generation and restructure the power sector to facilitate private participation, investment and competition.

Ultimately, we need a forward-looking Energy Ministry. One that appreciates better its legislated mandate to restore energy security and one which understands and supports the inexorable energy transition that is sweeping the world. We need to be ahead of the game, not dragging our feet. And embracing the energy transition will help us to restore energy security earlier, rather than later.

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